PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : GASEOUS STATE (Question Bank # 09)

SINGLE CORRECT

1. (D)

Sol. applied pressure by $N_2 = (740 - 40)$

$= 700$ torr

applied pressure by Ne = $(780 - 40)$

$= 740$ torr

$$\frac{P_f V_f}{RT} = \frac{P_{N_2} V_{N_2}}{RT} + \frac{P_{Ne} V_{Ne}}{RT}$$

$P_f(500) = 700 \times 250 + 740 \times 300$

$P_f = 794$ torr

total applied final pressure

$= 794 + 40 = 834$ mm of Hg

So nearest possible answer will be D option.

2. (A)

Sol. Rate of effusion $= \frac{PAN_A}{\sqrt{2\pi MR_T}}$

$$\frac{r_{He}}{r_{SO_2}} = \frac{P_{He}}{P_{SO_2}} \sqrt{\frac{M_{SO_2} \cdot T_{SO_2}}{M_{He} \cdot T_{He}}}$$

$$\frac{x}{n} = \frac{0.25}{1} \sqrt{\frac{64 \times 273}{4 \times 546}}$$

$$x = \frac{n}{\sqrt{2}}$$

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3. (B)

Sol. For velocity distribution

$$\frac{dN}{N} = 4\pi\left(\frac{m}{2\pi KT}\right)^{\frac{3}{2}} - e^{\frac{mv^2}{RKT}} u^2 du$$

$$\left[\frac{1}{2}mv^2 = K.E.\right] \qquad \text{(Let K.E. is } \varepsilon\text{)}$$

$$u = \sqrt{\frac{2\varepsilon}{m}} \qquad \text{Put in equation (i)}$$

$$\frac{dN}{N} = 4\pi\left(\frac{m}{2\pi KT}\right)^{\frac{3}{2}} - e^{\frac{E}{KT}} \frac{2\varepsilon}{m} \frac{1}{2\sqrt{\frac{2\varepsilon}{m}}} \cdot \frac{2}{m} d\varepsilon$$

$$\frac{dN}{N} = 4\pi\left(\frac{m}{2\pi KT}\right)^{\frac{3}{2}} - e^{\frac{E}{KT}} \cdot \sqrt{\frac{\varepsilon}{m}} \cdot \frac{\sqrt{2}}{m} d\varepsilon$$

$$\underbrace{\frac{1}{N}\frac{dN}{d\varepsilon}}_{\text{Say y}} = \underbrace{4\pi\left(\frac{1}{2\pi KT}\right)^{\frac{3}{2}}}_{\text{say constant}} - e^{\frac{\varepsilon}{KT}} \sqrt{\varepsilon} \qquad \text{..... (ii)}$$

for most probable K.E.

$$\frac{dy}{dE} = 0$$

$$\frac{dy}{dE} = \text{costant}\left(\sqrt{E}\, e^{-\frac{\varepsilon}{KT}}\left(-\frac{1}{KT}\right) + e^{-\frac{\varepsilon}{KT}} \frac{1}{2\sqrt{E}}\right) = 0$$

On solving : $\boxed{E = \frac{KT}{2}}$

4. (D)

Sol. for open container

$$n_1T_1 = n_2T_2$$

$$n_1(300) = \left(\frac{5}{8}n_1\right)(T_2)$$

$$T_2 = 480 \text{ kelvin}$$

$$T_2 = 207°C$$

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5. **(A)**

Sol. PM = dRT

(1) (30) = z (0.4) (0.0821) (773)

z = 1.1817

as z > 1 repulsive forces will be dominaht

6. **(B)**

Sol. $\left(P+\frac{an^2}{v^2}\right)v = nRT$

from ideal gas we know

PV = nRT

1 × 10 = nRT

Put value of nRT in equation (i)

$$\left(P+\frac{an^2}{v^2}\right)(v)=10$$

$$\left(0.96+\frac{a(0.5)^2}{10^2}\right)10=10$$

On solving a = 16 atm L^2/mol^2

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7. **(D)**

Sol. (A) From observance in gas A only repulsive forces are active so for this gas a = 0.

(B) From observances in gas B only attractive forces are active so for this gas b = 0.

(C) For gas C by getting minima and intersection point z = 1 we can form two equation by two equations two unknown a and b can be calculated.

(D) Clearly gas B have negative slope so this option must be wrong.

8. **(B)**

Sol. for real gas pv = znRT

$$\frac{P_2}{P_1}\cdot\frac{v_2}{v_1}=\frac{z_2}{z_1}\cdot\frac{T_2}{T_1}$$

$$\left(\frac{600}{300}\right)\left(\frac{v_2}{1}\right)=\frac{1.375}{1.072}\left(\frac{273}{473}\right)$$

$v_2 = 0.3701$ lit

$v_2 = 370.1$ ml

9. **(D)**

Sol. All the three statements are true so option will be D.

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MULTIPLE CORRECT

10. (B,C)

Sol. Initialy during compression pv = cosntant

<u>During compression pressure at valve-1 :</u>

$1 \times 60 \times 821 = P_1 \times 821 \times 45$

$P_1 = 1.33$ it is not grater then 1.5

So value-1 will not open during compression.

<u>During compression pressure at valve-2 :</u>

$1 \times 60 \times 821 = P_2 \times 821 \times 30$

$P_2 = 2$ atm

So value-2 will also not open during compression.

<u>During compression pressure at valve-3</u>

$1 \times 60 \times 821 = P_3 \times 20 \times 821$

$P_3 = 3$ atm it is greater then 2.5 atm

So valve-3 will open first till pressure of gas becomes equal to 2.5 atm.

<u>During compression pressure at valve-4</u>

$2.5 \times 20 \times 821 = P_4 \times 15 \times 821$

$P_4 = 3.33$ atm

So valve-4 will not open.

<u>During compression valve-5</u>

$2.5 \times 20 \times 821 = P_5 \times 10 \times 821$

$P_5 = 5$ atm it is grater then 4.8 atm

So valve-5 will open second after valve-3 till the pressure of gas beomes 4.8 atm.

(A) incorrect

(B*) Remaining mol of gas after opening the valve-3

$$2.5 \times \frac{821 \times 20}{1000} = n(0.821)(300)$$

$n = \frac{5}{3}$ so this option is correct

(C*) Correct

(D) Number of moles of gas at valve-5 will not be zero as amount of gas is not becomes zero after opening valve-5

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11. (A,C,D)

Sol. Initial moles of $NH_3 \Rightarrow$

$$\left(\frac{3}{2}\right)(4.92) = n_{NH_3}(0.0821)(300)$$

$$n_{NH_3} = \frac{3}{10}\text{mole}$$

at $t = 0$ when $n_{NH_3} = \frac{3}{10}$, height difference is 38 c.m.

Now NH_3 will react and when become $\frac{1}{2}$atm then difference in levels becomes 38 c.m.

So mole of NH_3 when pressure becomes $\frac{1}{2}$atm

$$\left(\frac{1}{2}\right)(4.92) = n_{NH_3}(0.0821)(300)$$

$$n_{NH_3} = \frac{1}{10}\text{mol}$$

So reacted mol of NH_3 are $\frac{3}{10} - \frac{1}{10} = \frac{2}{10}$mol

so in this used mol of $H_2S = \frac{1}{2}\left(\frac{2}{10}\right) = \frac{1}{10}$mol

time require for 0.1 mol H_2S will be

$$= \frac{0.1}{0.01} = 10\,\text{second}$$

Now NH_3 will completely react and H_2S start accumulating in flask.

Now 38 c.m. difference in level will be when pressure applied by H_2S becomes $\frac{1}{2}$atm.

$$\left(\frac{1}{2}\right)(4.92) = n(0.0821)(300)$$

$$n = \frac{1}{10}\text{mol}$$

mole of H_2S consumed in reaction with $NH_3 = \frac{3}{20}$atm

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total used mole of $H_2S = \frac{1}{10} + \frac{3}{20} = \frac{5}{20} = \frac{1}{4}$

time require for $\frac{1}{4}$ mole H_2S will be

$= \frac{\frac{1}{4}}{0.01} = 25\,\text{sec}$

again difference of 38 cm. will create when pressure becomes $\frac{3}{2}$ atm due to H_2S

$\left(\frac{3}{2}\right)(4.92) = n(0.0821)(300)$

$n = \frac{3}{10}$

mole of H_2S cousumed in reaction with $NH_3 = \frac{3}{20}$ mol

total used mole of $H_2S = \frac{3}{10} + \frac{3}{20} = \frac{9}{20}$

time require for $\frac{9}{20}$ mole of H_2S

$\Rightarrow \frac{\frac{9}{20}}{0.01} = 45\,\text{sec}.$

12. **(A,B,C)**

Sol. (A) $\frac{P_{A_4}}{P_{n-1}} = \frac{P/5}{P/n} = 3$

$n = 15$

(B) $\frac{P_{A_4}}{P_{n-1}} = \frac{P/4}{P/n-1} = 3$

$n = 13$

(C) $\frac{P_{A_3}}{P_{A_5}} = \frac{P/4}{P/6} = \frac{6}{4} = 1.5$

(D) $\frac{P_{A_6}}{P_{A_1}} = \frac{P/6}{P/1} = \frac{1}{6}$

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13. **(C)**

Sol. $PV_m = 28$ at 0°N

$PV_m = 40$ at 100°N

Let T kelvin = T°N + x

so ideal gas equation becomes

$PV_m = R(T°N + x)$

$40 = R(100 + x)$ ..... (i)

$28 = R(0 + x)$ ..... (ii)

(i) ÷ (ii)

$$\frac{40}{28} = \frac{100 + x}{x}$$

so value of 0 K will be equal to – 233.33°N

14. **(B)**

Sol. $28 = R(0 + x)$

$28 = R(233.33)$

$R = 0.12$ atm. lit/mol-K

15. **(A)**

Sol. $Pv = nR\,(T°N + 233.33)$

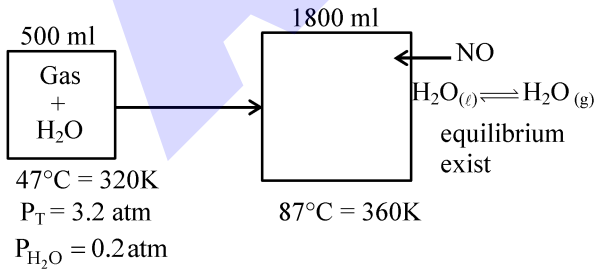
$2 \times v = 2 \times 0.12\,(66.67 + 233.33)$

$v = 36$ litre

SECTION - II INTEGER

1. **(1.00)**

Sol.



$$\frac{3.2 \times 500}{320} = \frac{P_2 \times 1800}{360}$$

$$P_2 = \frac{3.5 \times 5}{16} = 1\,\text{atm}$$

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2. (8.00)

Sol. Let initial moles of

$n_{NH_3} = x$

$n_{CO_2} = y$

$n_{N_2} = z$

Condition on I[st] bulb :-

$60 \times 12.315 = (x + y + z)\ (24.63)$

$30 = x + y + z$ ....(i)

After opening valve-I condition on I[st] and II[nd] bulb :-

Pressure due to CO_2 & N_2 after opening I[st] valve

$= 17 - \text{v.p. of } NH_3$

$= 16$ atm

On bulb-I

$16 \times 12.315 = (y - y' + z - z')\ (24.63)$

here y' & z' are gaseous moles of CO_2 and N_2 that transfermed to 2[nd] bulb.

$$y' = \frac{3y}{5} \qquad z' = \frac{3z}{5}$$

$$16 \times 12.315 = \left(y - \frac{3y}{5} + z - \frac{3z}{5}\right)(24.63)$$

$$8 = \frac{2y}{5} + \frac{2z}{5}$$

$20 = y + z$ ..... (ii)

after opening valve-II

Condition on 1[st], 2[nd] and 3[rd] bulb

In this situation NH_3 and CO_2 will completely liquify only N_2 will remain in system

Let moles of N_2 in second and third bulb are z' and z'' so.

On bulb-I

$4 \times 12.315 = (z - z' - z'')\ 24.63$

$2 = z - z' - z''$ ....(iii)

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as we know pressure in all three bulbs will be equal 50 and volume is already same

So $nT = \text{constant}$

$(z - z' - z'')300 = z'(200) = z''(100)$ ....(iv)

On solving $z = 11$ mole

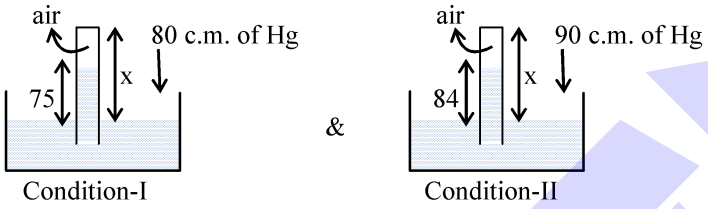
So $y = 9$ mole

and $x = 10$ mole

So, value of $x + y - z = 10 + 9 - 11 = 8$

3. (4.00)

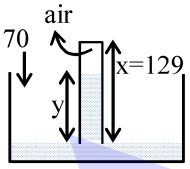
Sol.



$P_1v_1 = P_2v_2$ for air

$5(x - 75) = 6(x - 84)$

$x = 129$ c.m.



$5(129 - 75) = (70 - y)(129 - y)$

$y^2 - 199y + 8760 = 0$

On solving

$y = 133.5, 65.5$

133.5 will be not possible because y can not be more then 129.50

$y = 65.5$

Ans. $\dfrac{8(65.5)}{131} = 4$

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MATRIX MATCH TYPE

1. (A-P,S; B-R, C-P,Q, D-R)

(A) Hydrogen gas P = 200 atm, T = 273K

gas will behave as real gas and repulsive forces are dominating because for H_2, a = 0 So options will be $\Rightarrow$ P, S

(B) Hydrogen gas (P ~ 0, T = 273K)

as $P \to 0$ So $v_m \to \infty$ so gas will behave jsut like ideal gas $\Rightarrow$ R

(C) CO_2 (P = 1 atm, T = 273K)

CO_2 will be have as real gas in lower pressure region so attractive forces are dominating $\Rightarrow$ P, Q

(D) Real gas with very large volume

$V \to \infty$

So real gas will be have like ideal gas $\Rightarrow$ R

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